UNRAMIFIED FINITE SEPARABLE EXTENSIONS OF A DISCRETELY VALUED FIELD ARE CLOSED UNDER SUBEXTENSION, GALOIS CLOSURE AND COMPOSITUM

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Editorial note. The statement contains the author's duplicated article in 'there exists a a finite separable extension', and the proof ends with 'Details omitted.'; both are reproduced unchanged. The nine references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The integral closure of a discrete valuation ring in a finite separable extension and multiplicativity of ramification indices. used at proof lines 33089-33096. Is the structure that lets the unramifiedness assumption be transported to a subextension.
- Unramifiedness with respect to A as etaleness of A -> B. used at proof lines 33112-33115. Converts the hypothesis into a form to which permanence of etale ring maps applies.
- The normal closure as a Galois extension receiving a surjection from a tensor power. used at proof lines 33102-33110. Produces the Galois extension required in part (2).
- Permanence of etale ring maps under tensor products, and etale extensions of a discrete valuation ring as products of Dedekind domains. used at proof lines 33116-33123. Is the step showing the tensor power is a product of Dedekind domains, from which unramifiedness follows.

Source reference. Original lemma 00U6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1100–1100. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BMF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14628–14628. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EXM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18662–18662. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EXL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18661–18661. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 10881: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10881–10881. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32952: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32952–32952. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33006: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33006–33006. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K.*

1. *If $M/L/K$ are finite separable extensions and M is unramified with respect to A, then L is unramified with respect to A.*
2. *If L/K is a finite separable extension which is unramified with respect to A, then there exists a Galois extension M/K containing L which is unramified with respect to A.*
3. *If L_1/K, L_2/K are finite separable extensions which are unramified with respect to A, then there exists a a finite separable extension L/K which is unramified with respect to A containing L_1 and L_2.*

Proof. We will use the results of the discussion in Remark 33006 without further mention.

Proof of (1). Let $C/B/A$ be the integral closures of A in $M/L/K$. Since C is a finite ring extension of B, we see that $\mathrm{Spec}(C) \to \mathrm{Spec}(B)$ is surjective. Hence for ever maximal ideal $\mathfrak{m} \subset B$ there is a maximal ideal $\mathfrak{m}' \subset C$ lying over $\mathfrak{m}$. By the multiplicativity of ramification indices (Lemma 32952) and the assumption, we conclude that the ramification index of $B_{\mathfrak{m}}$ over A is 1. Since $\kappa(\mathfrak{m}')/\kappa_A$ is finite separable, the same is true for $\kappa(\mathfrak{m})/\kappa_A$.

Proof of (2). Let M be the normal closure of L over K, see Fields, Definition 0BMF. Then M/K is Galois by Fields, Lemma 0EXM. On the other hand, there is a surjection

$$L \otimes_K \ldots \otimes_K L \longrightarrow M$$

of K-algebras, see Fields, Lemma 0EXL. Let B be the integral closure of A in L as in Remark 33006. The condition that L is unramified with respect to A exactly

means that $A \to B$ is an étale ring map, see Algebra, Lemma 00U6. By permanence properties of étale ring maps we see that

$$B \otimes_A \ldots \otimes_A B$$

is étale over A, see Algebra, Lemma 00U2. Hence the displayed ring is a product of Dedekind domains, see Lemma 10881. We conclude that M is the fraction field of a Dedekind domain finite étale over A. This means that M is unramified with respect to A as desired.

Proof of (3). Let $B_i \subset L_i$ be the integral closure of A. Argue in the same manner as above to show that $B_1 \otimes_A B_2$ is finite étale over A. Details omitted. □